

Feature Selection Strategy for Classification of Single-Trial EEG Elicited by Motor Imagery

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Abstract—Brain-Computer Interface (BCI) provides new means of communication for people with motor disabilities by utilizing electroencephalographic activity. Selection of features from Electroencephalogram (EEG) signals for classification plays a key part in the development of BCI systems. In this paper, we present a feature selection strategy consisting of channel selection by fisher ratio analysis in the frequency domain and time segment selection by visual inspection in time domain. The proposed strategy achieves an absolute improvement of 7.5% in the misclassification rate as compared with the baseline system that uses wavelet coefficients as features and support vector machine (SVM) as classifier.

Keywords- Brain-computer interface, feature selection, EEG classification, discrete wavelet transform, support vector machine.

I. INTRODUCTION

Brain-Computer Interface (BCI) is a system which connects humans with computers not through the conventional way but through brain signals recorded by devices like EEG and Electroencephalogram (EECoG). EEG is currently the most common method employed for BCI research because of its non-invasive nature and its inexpensiveness [1, 2]. It has been observed that the imagination of a particular task shows discriminating changes in the EEG signal. BCI uses this phenomenon for communication between humans and computers [3]. People suffering from neuro-muscular diseases have problems with communication and movement related tasks. As a consequence, they have difficulty in interacting with and expressing their intentions and feelings to others. As such, they are in a locked-in state, where they can see and feel but cannot move any of their muscles nor speak to express themselves. However, their hearing, vision and reasoning are normal. BCI research tries to help these people by understanding their intentions through EEG signals. If one can decipher the various signals coming from the brain of a person with neuro-muscular diseases and correlate it with various activities with a high degree of accuracy, it will be a great help for the person.

Two important parts in BCI research are the extraction and selection of features in the brain potentials elicited by imaginary tasks or external stimuli. Since some channels may

be noisy and irrelevant for a certain activity and the inclusion of such channels will degrade the BCI performance, it is very important to select an optimal and relevant channel set. Similarly, time segment selection from the entire signal is another important step in improving the performance. Channel and time segment selection has been previously studied in the literature. Channel selection is done by utilizing the support vector machines in [4] and by using genetic algorithms in [5]. In [6], four data-segment related parameters, namely, length of data segment (LDS), the starting position of the data segment, AR order and number of trials (NT) were selected to study its effect on feature extraction and classification. The dataset which is used in this study represents motor imagery tasks from the same limb. On this dataset, identification of task parameters from movement related cortical potentials has been studied, where limited effort has been put in investigating the effects of channel and time segment selection on the classification accuracy [7]. Therefore, this study focuses on a feature selection strategy to reduce misclassification rate, including channel selection by fisher ratio analysis in the frequency domain and time segment selection by visual inspection. The effect of these strategies on the misclassification rate is analyzed and discussed.

The rest of the paper is organized as follows: Section II describes the data, feature selection and extraction strategy and the classification method. In section III the baseline system is presented and the experimental results are discussed. The paper is concluded in section 4.

II. DATA AND METHODS

This section briefly describes the dataset used in this work and then presents the methods for feature extraction, feature selection and classification.

A. Data

Electrodes mounted in a standard EEG Cap (Electro-Cap International, 32 channels) according to the 10-20 system is used for the EEG recording [7]. Subjects were asked to

imagine isometric plantar flexion of the right foot at different target torque (TT) and Rate of torque development (RTD). The Maximal Voluntary Contraction Torque (MVCT) is defined as the torque exerted for the maximal contraction of the isometric plantar flexion and is calculated as the average torque of three maximal contractions. Depending upon the MVCT, TT can be either “low” or “high”. 30% of MVCT is considered low and 60% of MVCT is considered high in this study. RTD can also be “ballistic” or “moderate” depending upon how fast the TT is reached. Reaching TT ‘as fast as possible’ is considered ballistic and in 4s is considered moderate. Subjects were given visual cues on a computer screen instructing them on which task to perform and when it should be performed. The tasks were divided into four types:

- BH: Ballistic movement to High torque.
- BL: Ballistic movement to Low torque.
- MH: Moderate movement to High torque.
- ML: Moderate movement to Low torque.

For detailed description of the task and data collection process, refer [7]. The present study used only two classes: BH and BL, and EEG recordings from 6 subjects.

B. Time Segment Selection

The averages of all the single-trials of both classes, as depicted in Fig. 1, were visually inspected for all channels. Conclusions regarding the best discriminating subset from the total signal set were drawn from this inspection. The following two EEG segments were chosen for further analysis:

- 2s segment is extracted from 1s prior to 1s after onset of the task, and
- 2s segment is extracted from 2 seconds after the onset of the task.

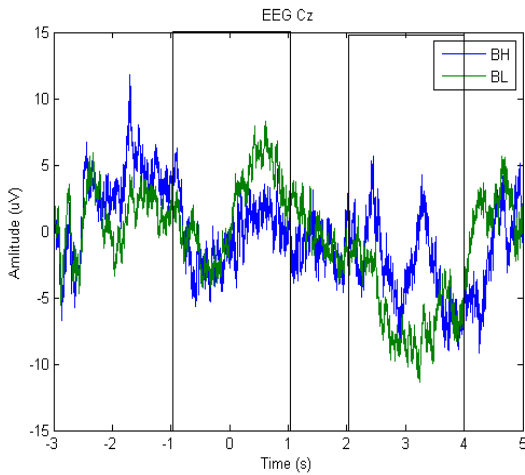


Figure 1: BH and BL class graph of sub F for channel Cz. Onset of the task is at 0s, and the two segments extracted are from -1s to 1s and from 2s to 4s.

C. Feature Extraction

The segmented EEG trials obtained from the time segment selection step were band-pass filtered between 7-30Hz and then processed using Discrete Wavelet Transform (DWT).

In the multi-resolution analysis framework with orthogonal wavelet, DWT maps the signal into scaling function $\phi(t)$ and the corresponding mother wavelet function $\Psi(t)$. They are related to the low-pass and high-pass filter $h(n)$ and $g(n)$, respectively by equations (1) and (2) and $\Psi(t)$ can be derived from the scaling filter itself as well [8]:

$$\phi(t) = \sqrt{2} \sum_n h(n) \phi(t - n) \quad (1)$$

$$\Psi(t) = \sqrt{2} \sum_n g(n) \phi(t - n) \quad (2)$$

To select the optimal mother wavelet for feature extraction, a set of 21 $h(n)$ filters of length 4 was generated, by parameterizing them with a value in the range from $-\pi$ to π , as described in [9]. With mother wavelet Ψ , the DWT gives a set of coefficients $d_x(j, k)$, which are partly localized in time and frequency. The marginals $m_x(j)$ defined in (3) were used as features in order to make the decomposition insensitive to time alignment [9].

$$m_x(j) = \sum_{k=0}^{N/2^j-1} c_x(j, k) \quad j = 1, \dots, J \quad (3)$$

$$c_x(j, k) = \frac{|d_x(j, k)|}{\sum_{j=1}^J \sum_{k=0}^{N/2^j-1} |d_x(j, k)|} \quad (4)$$

where, J is the deepest level of decomposition and is equal to $\log_2 N$, N is the length of the signal x , $d_x(j, k) = \langle x(t), \Psi_{j,k}(t) \rangle$, j is related to scale and k is related to translation by $\Psi_{j,k} = 2^{-j/2} \Psi(2^{-j}t - k)$

Once the marginals of all detail spaces were calculated for all EEG channels and all 21 mother wavelets, each of the 21 feature spaces corresponding to each of the mother wavelets were separately classified using a SVM, which is described in subsection II-E. Only the feature space corresponding to the mother wavelet with lowest misclassification rate was used in the subsequent analysis.

D. Channel Selection

The discrete wavelet transform (DWT) of the signal from each channel is used to find out the degree of difference of each channel data (feature) from one class to another class using the fisher ratio analysis [10]. Depending upon this, an optimal subset of channels for all the subjects are found for further analysis.

The fisher ratio between classes BH and BL for the i^{th} channel is given by the following equation:

$$fr_i = \frac{(m_{BH_i} - m_{BL_i})^2}{\sigma_{BH_i}^2 - \sigma_{BL_i}^2} \quad (5)$$

where m_{BH_i} , m_{BL_i} and σ_{BH_i} , σ_{BL_i} are the mean and variance of the samples belonging to BH and BL, respectively.

Four subsets of channels from the total 32 channels were chosen depending upon the fisher ratio calculation, best 9 channels, best 8 channels, best 7 channels and best 6 channels. These 4 feature spaces were classified using the classification method described in the next subsection.

E. Classification

Support vector machine is used for the classification process since it has good generalisation property, does not suffer from the curse of dimensionality and is insensitive to overtraining [11]. Classification normally involves separating the whole data set into training and testing set. Each instance of the training set has a feature set and a corresponding class label. By using the training set, an SVM model is created to predict the class information of the test data. If a training data vector is represented by x_i where $x_i \in R^n$ and its corresponding class is represented by $y_i \in (-1,1)$, SVM requires the solution of the following optimization problem [12]:

$$\min_{w, b, \xi} \quad \frac{1}{2}(w^T w) + C(\sum_{i=1}^l \xi_i) \quad (6)$$

$$\text{subject to } y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i \quad (7)$$

$$\xi_i \geq 0, \quad i=1, \dots, l$$

Here x_i are mapped into a higher dimensional space by the function ϕ , ξ_i is a slack variable and $C > 0$ is the penalty parameter of the error term, w is the weight vector, and b is the bias of hyperplane. A Gaussian kernel function and a 3-fold cross validation were used in this study.

III. EXPERIMENTS AND RESULTS

The baseline system was developed based on the study described in [7], which consists of band pass filtering of the

EEG signals in the range 0.1 to 75Hz and notch filtering at 50Hz. Features were extracted using DWT as described in subsection II-C. Seven pre-selected channels namely, F3, F4, C3, Cz, C4, P3 and P4 were used together for the SVM based classification.

The proposed feature selection strategy was applied to each of the 6 subject's data, namely, sub A, B, C, D, E and F, separately. Four subsets of the channels from a total of 32 channels were individually used for classification in the SVM. The four channel subsets, namely, best 6, best 7, best 8 and best 9 were selected based on the fisher ratio calculation. The misclassification rates of all the subjects using the proposed method, with the four subsets of channels, all channels, and the 7 channels used in the baseline system are displayed in Table 1 together with the results of the baseline system.

The comparison of the results clearly shows that the average misclassification rate of all the subjects with the proposed feature selection strategy outperforms the baseline system in which the feature selection strategy presented in this study is not used. An absolute improvement of 7.5% in the misclassification rate with the best 7 channels selected using fisher ratio is observed as compared with the baseline system.

Comparison of the average of misclassification rate of all the subjects between the four optimal subset of channels, i.e., "best 9, best 8, best 7 and best 6" shows minimal differences in the misclassification rate. On the other hand, when all the channels are used for classification, the misclassification rate declined by 5.52% from the best 7 channels. This indicates the necessity of feature selection prior to the classification process for good performance.

Furthermore, selection of channels without employing the fisher ratio analysis i.e. using the fixed 7 channels used in the baseline system for classification shows a decline of 4.79% in the average misclassification rate from the best 7 channels selected using fisher ratio analysis. This shows that channel selection based on the degree of separability between two classes, measured using fisher ratio improves misclassification accuracy.

Table 1 Misclassification rate (%) for all the subjects

	Proposed system with feature selection strategy						Baseline System
	best 9 channels	best 8 channels	best 7 channels	best 6 channels	7 Channels used in the baseline system	All channels	
sub A	13.63	12.12	12.2	9.09	28.78	24.24	33.33
sub B	44.79	42.7	44.79	48.95	50	50	40.62
sub C	2.56	1.28	1.28	0	6.41	2.56	8.97
sub D	0	0	0	2.77	0	0	8.33
sub E	15.47	16.66	16.66	13.09	16.66	16.67	22.61
sub F	33.33	31.25	27.08	33.33	27.08	41.67	33.33
Average	18.30±17.55	17.34±16.86	17.00±16.9	17.87±19.24	21.49±17.94	22.52±20.32	24.53±13.57

IV. CONCLUSION

This study focuses on the feature selection from the EEG signal for the motor imagery tasks from the same limb. Two stages of feature selection were employed here: channel selection and time segment selection based on fisher ratio and visual inspection of the data, respectively. Its effects on the classification accuracy were studied and the results suggest that smaller and relevant subsets of channels achieve better performance than a large number of channels which may include irrelevant and noisy channels. This also reduces the complexity of the whole BCI system, therefore achieving good overall performance.

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